DESCRIPTION

The 24G radar board RKB1201T is a Traffic radar front-end demo board, designed with the high performance radar chip SRK1201L. It has one transmitter and two receivers. This board also contains ADF4159 PLL for option.

FEATURES

- TX frequency range: 24~24.25GHz.
- TX EIRP 28dBm, with 18dB control range, 0.5dB step.
- Phase noise: 103dBc/Hz@1MHz
- RX noise figure: 9 dB DSB typical.
- RX quadrature phase imbalance <5deg
- RX quadrature amplitude imbalance <1.5dB.
- RX MAX gain 50dB, with 16dB control range, 1.5dB step.
- 5~3.3V power supply, Current: 280mA MAX, 0.9W MAX

APPLICATIONS

- Traffic Monitoring
- Industrial Applications

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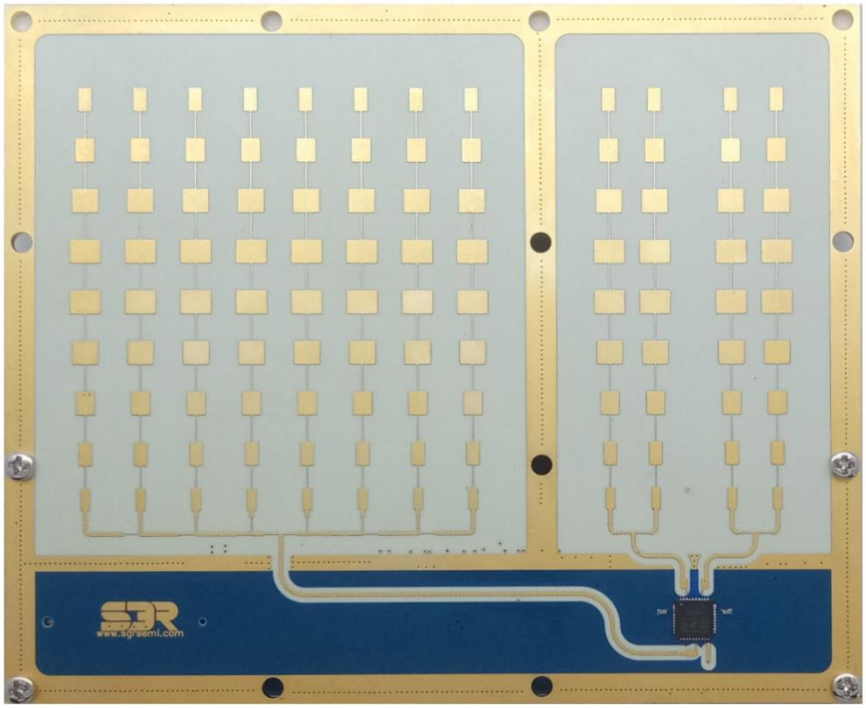

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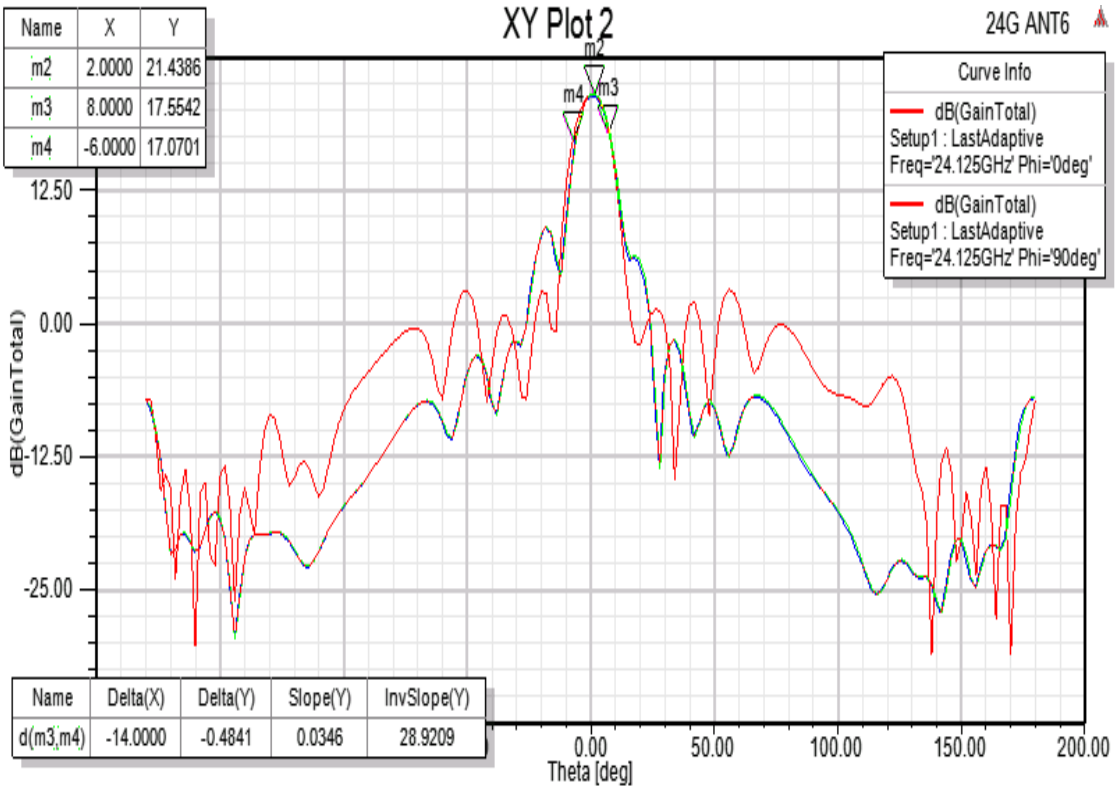
ELECTRICAL CHARACTERISTICS

Parameter	MIN	TYP	MAX	UNITS	CONDITIONS
Power supply					
DCC	3.3	3.5	5	V	
Current			280	mA	All function on
Current (Chip disable)			28	mA	PLL and SOC disable
Transmitter					
Tx frequency range	24		24.25	GHz	
Output power EIRP			28	dBm	
Phase noise		103		dBc/Hz	@1MHz
Tx power control range		18		dB	
Tx power control step		0.5		dB	
Receiver					
noise figure		9		dB	DSB
quadrature phase imbalance		1	5	deg	
quadrature amplitude imbalance		1	1.5	dB	
IF gain control range	34		50	dB	
IF gain control step		1.5		dB	
IF lower cutoff frequency		50		Hz	
IF upper cutoff frequency		100		KHz	
DC-Offset		1.5		V	
Antenna					
TX antenna gain		21		dB	
TX side lobe suppression		18		dB	
RX antenna gain		16		dB	
RX side lobe suppression		18		dB	
TX antenna pattern (3dB)		±7		deg	azimuth
		±7		deg	elevation
RX antenna pattern (3dB)		±26		deg	azimuth
		±7		deg	elevation

ANTENNA PATTERN

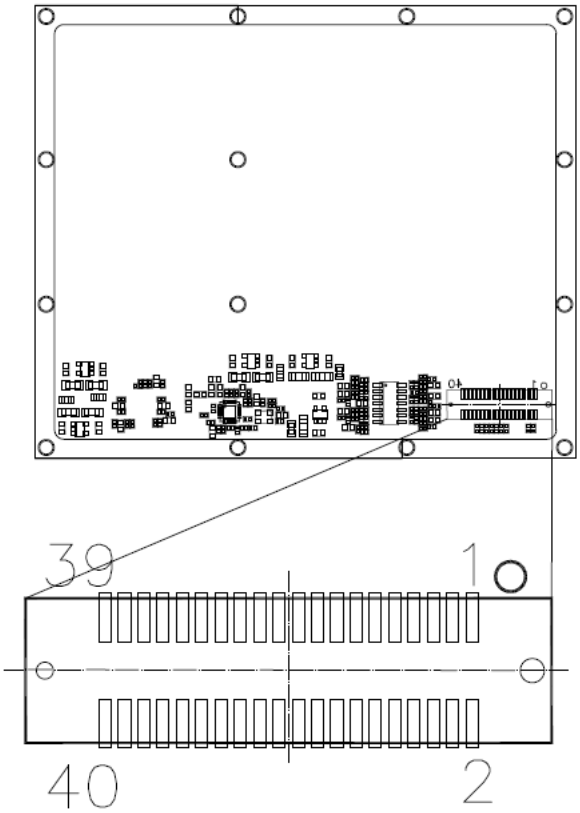
RX Antenna pattern：

TX Antenna pattern：



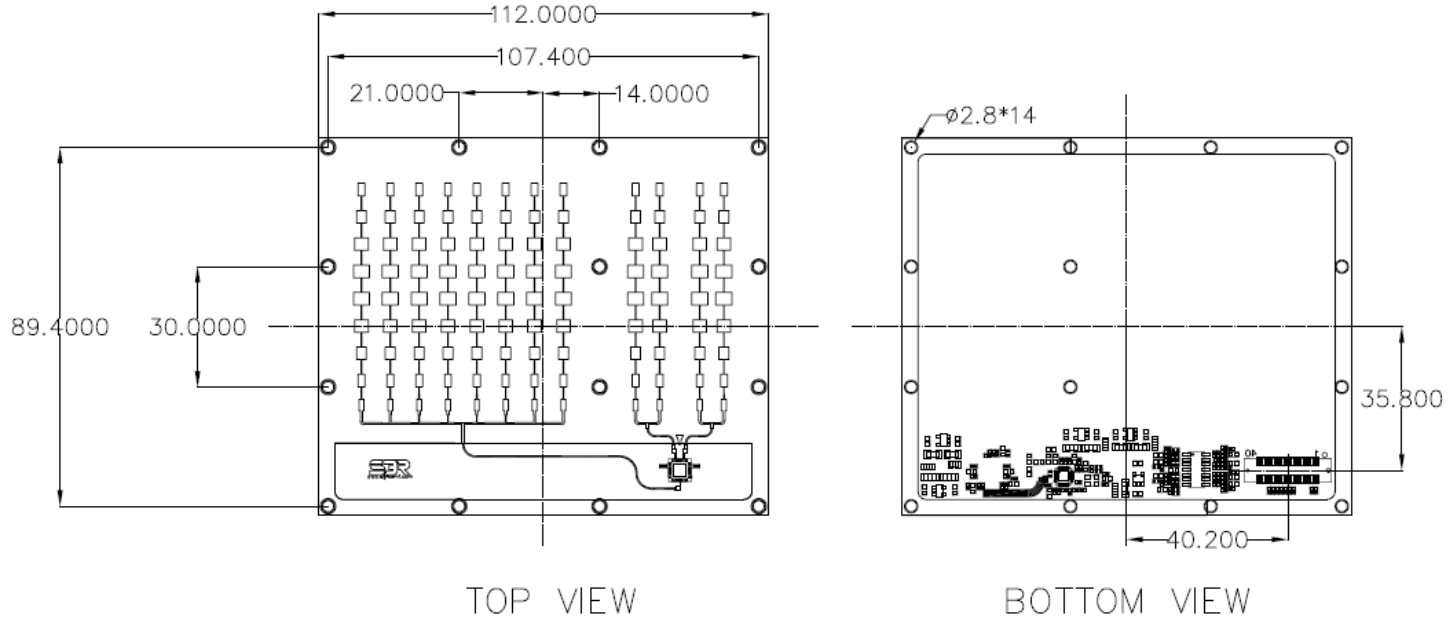
CONNECTOR INFORMATION

The connector P/N is 5179030-1, mating connector P/N is 5177983-1:



pin	Name	notes
1 Control and Power connector		
1,2,4,5,7,8,9,12,13,15,16,17, 20,21,23,24,25,28,29,31,32, 33,35,36,37,39	GND	
34	VT	SOC SRK1201 VT pin
38,40	VDD	3.5V
6	TXD	PLL ADF4159 TXDATA pin
22	LEP	SPI ADF4159 LE
10	MUX	PLL ADF4159 MUX pin
30	LER	SPI SRK1201 LE
26	CLK	SPI CLK
18	DAT	SPI DAT
14	ANA	SOC SRK1201 ANA pin
19	IFI1	IF Signal Channel 1, I
27	IFQ1	IF Signal Channel 1, Q
3	IFQ2	IF Signal Channel 2, Q
11	IFI2	IF Signal Channel 2, I

MECHANICAL OUTLINES



MORE INFORMATION

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2017/9/8 Rev. 1

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